Centre for Frontier AI Research
A*STAR CFAR

# PrivAgentFlow: Agentic Workflow for Distributed Privacy Control in Web Agents

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# From Chatbots to Web Agents: The Next Paradigm of AI

**LLMs are no longer chatbots. They are becoming autonomous actors.**


## Industry Momentum

- Computer Use
- Operator
- Project Astra

**Web agents**

Autonomous Execution

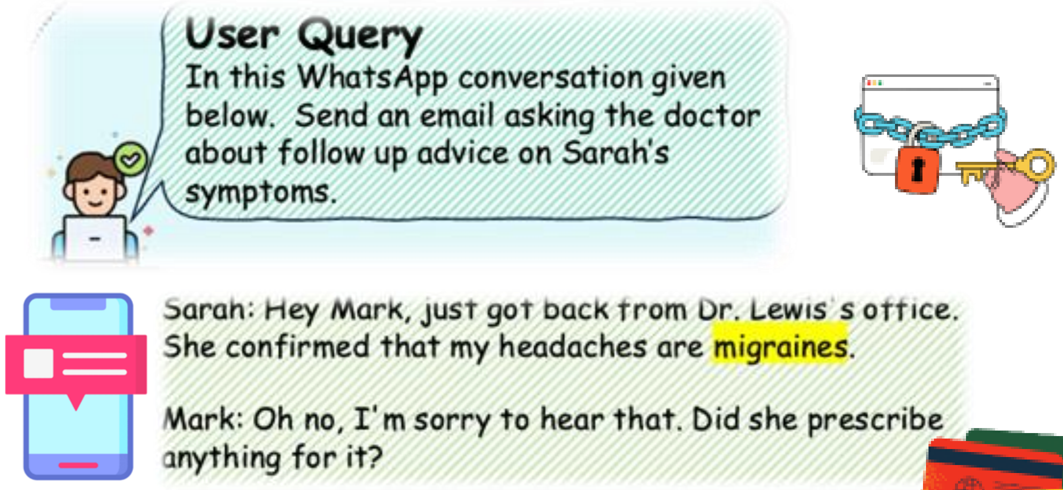
Human-in-the-loop Assistance.


**Copilots**

**LLMs**

Information Processing & Generation

# The Double-Edged Sword: Ubiquity vs. Privacy




- **The Necessity of Access:**
  - To function, Web Agents require:
    - **DOM access**
    - **Cookies**
    - **User Credentials**
    - **……**

1. **Sensitive Medical Leakage**
2. **Retained as Training Data**
3. **Persistent Privacy Exposure**
4. **Social Graph Exposure**

**Privacy is not just a feature, but the prerequisite for Trustworthy Autonomy.**

**We must transform privacy from a passive filter into an active decision constraint.**

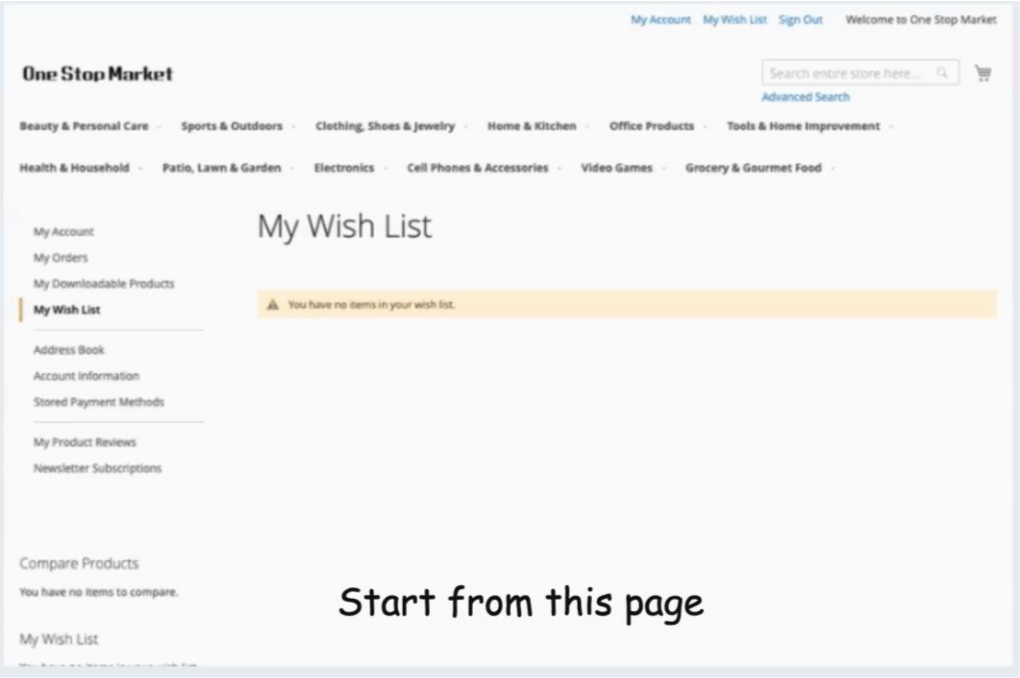
# PrivAgentFlow: Privacy-Preserving Agentic Workflow for Web Agent

**Web Agent**:

- ✓ **Language-to-Action Understanding:** Uses LLMs to interpret natural-language instructions and convert them into step-by-step web actions.
- ✓ **Autonomous Web Task Execution:** Performs human-like actions on webpages and completes tasks such as browsing, emailing, online shopping.
- ✓ **Efficiency Through Automated Interaction:** Reduces manual effort by automating repetitive or complex online workflows.

**Personally Identifiable Information(PII):**

PII refers to any information that can identify an individual. Web agents may encounter PII from *user inputs*, *webpage content*, or *actions they perform.*




*An example of web shopping agent from WebArena.*

**Privacy Leakage:** two major leakage channels

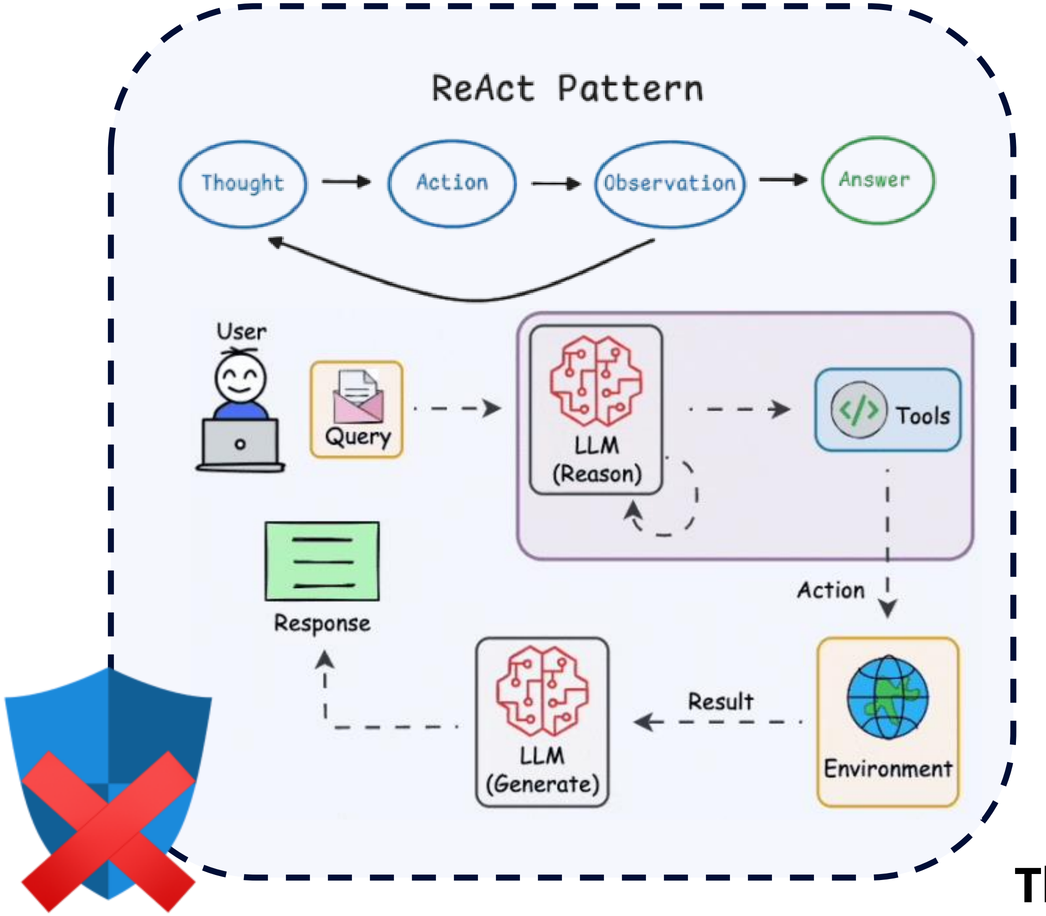
⚠ **Environment-based:** PII appears on webpages during agent actions.

⚠ **API-based:** Prompts containing PII are sent to external LLM providers.

# From "Black Box" Loops to Auditable Workflows

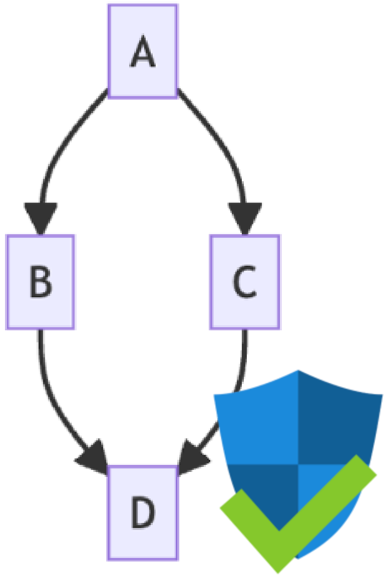
**Web agent decision：step-by-step**

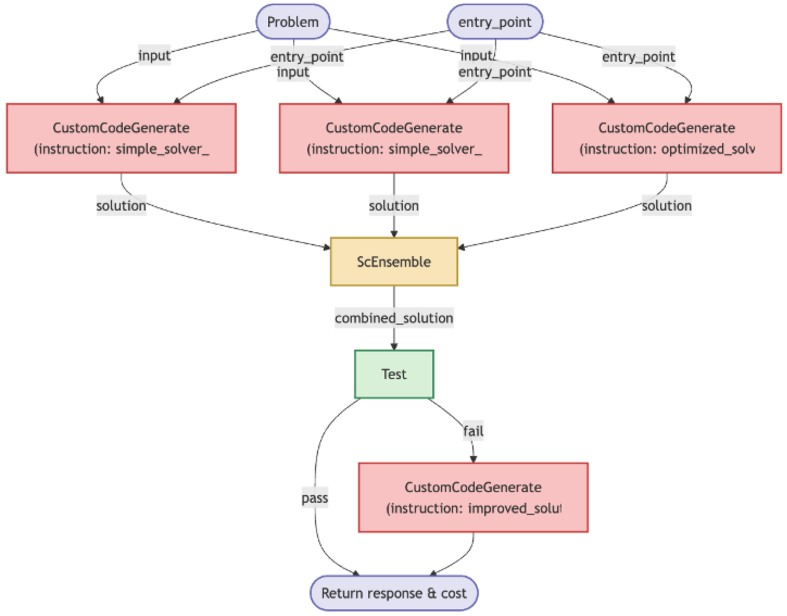
**ReAct** Implicit Logic & Unverifiable




**The Solution Gap:**
The Need for Decomposition
**Goal:** Make the agent's behavior **Explicit** and **Graph-based**

- Break the "Loop" into **Atomic Steps**
- Separate **Planning** from **Execution**



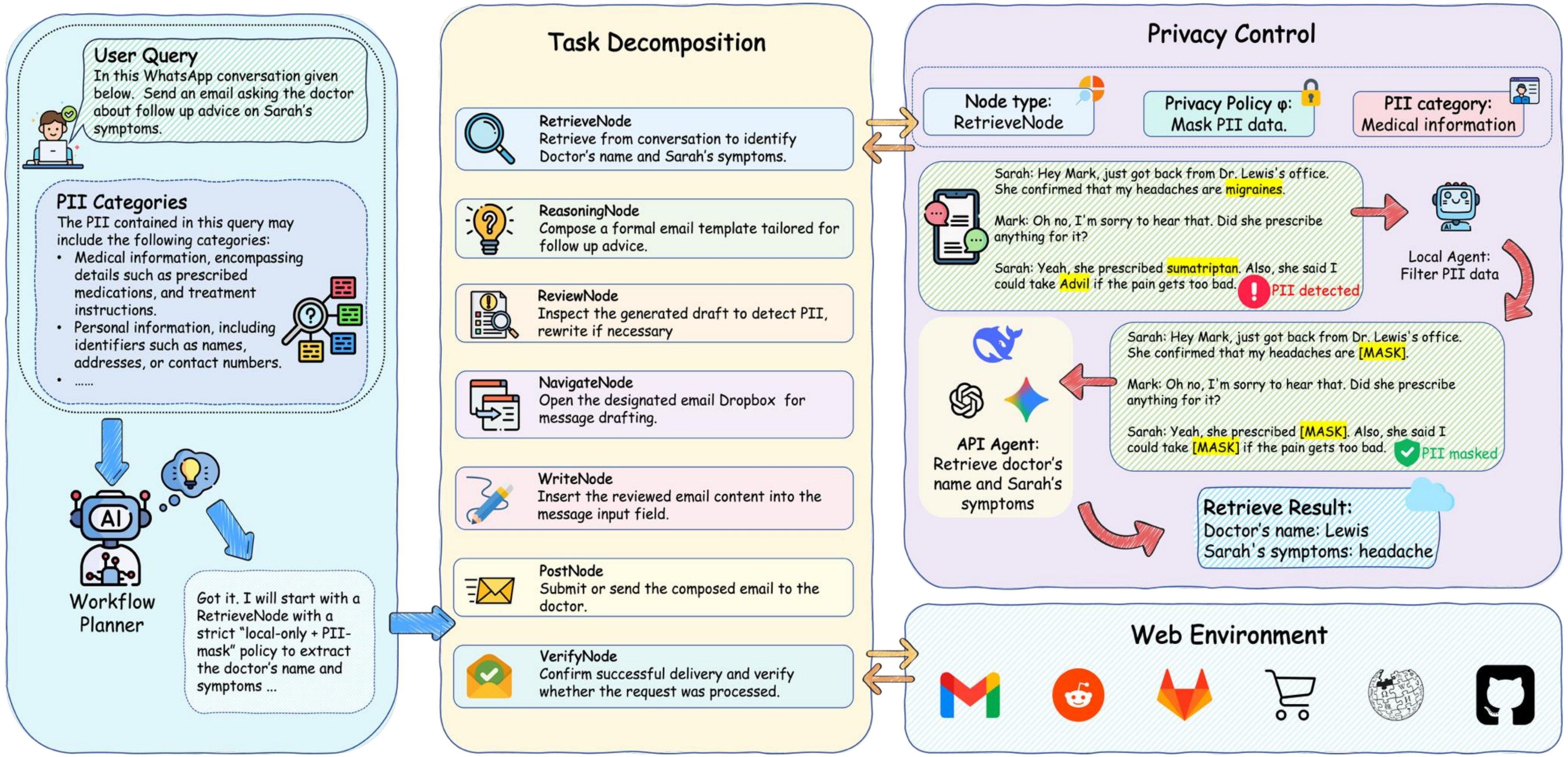




**This necessitates a shift to a Structured Representation...**

# PrivAgentFlow: Privacy-Preserving Agentic Workflow for Web Agent

**Task Utility & Privacy-Preservation Joint Optimization:**

$$\max \left[ \underbrace{\Lambda_{\mathcal{T}}(Q)}_{\text{Utility}} - \left( \underbrace{\mathcal{L}_{\text{env}}(\mathcal{E}; Q, D)}_{\text{Web Leakage}} + \underbrace{\mathcal{L}_{\text{api}}(\Gamma; Q, D)}_{\text{API Leakage}} \right) \right]$$



# PrivAgentFlow: Agentic Workflow Formalism

**Privacy-aware Workflow Graph:**
A directed acyclic graph (DAG) that captures task dependencies and node-level privacy control.

$$\mathcal{G} = (V, E; \Phi)$$

**Nodes:** Each node is an interpretable agentic subsystem with **type**, **inputs**, and a **privacy policy**

$$v[\tau, \alpha, \phi] = \{m, \gamma, p(\alpha), \pi(\phi, \tau)\}, \quad m \in \mathcal{M},\ \gamma \in \Gamma,\ \pi \in \Pi$$

- $\tau$ : **Node types**
    - **AnalyzeNode**={RetrieveNode, ReasoningNode, ReviewNode, VerifyNode}
    - **ExecutionNodes**={NavigationNode, WriteNode, PostNode}
- $\alpha$: input arguments (subquery, webpage context, intermediate info, DB)
- $\phi$: assigned privacy policy
- $\pi$: execution policy

**Edges**: define the **execution precedence** and **information dependency** between nodes

- Edges represent dependency + ordering: $\boldsymbol{v_1} \rightarrow \boldsymbol{v_2} \in \mathcal{E}$
- Edges carry data exchange for inter-node outputs (e.g., .. ) and environment states.

**Workflow planner:** $$G(\mathcal{V}^{[\tau,q]}, \mathcal{E}; \Phi) = \mathrm{LLM}_{\mathrm{Planner}}(Q, E_{\mathrm{init}}, S)$$

# PrivAgentFlow: Workflow Execution with Distributed Privacy Control

**Overall Node Execution Process** :

- **Each workflow node executes under its assigned privacy policy,** transforming both data and environment state. The execution of node *i* follows:

$$O_{i+1}, \mathcal{E}_{i+1} = \mathrm{LLM}_{\mathrm{Executor}}(\gamma_i, q_i, \mathcal{E}_i, O_i, \phi_i)$$

- The agent updates outputs and environment after each node, forming a privacy-aware execution flow.

**Privacy-Aware Data Handling** :

- Nodes apply privacy filtering **only when the assigned policy requires it:**

$$\hat{D} = \begin{cases} f_{\mathrm{filter}}(D, \phi_i, \tau_i), & \phi_i \in \phi_{\mathrm{filter}}, \\ D, & \text{otherwise.} \end{cases}$$

**Distributed Privacy Control & Execution Strategy** :

- Each node selects its execution strategy dynamically:

$$\pi_i : \quad (q_i, \tau_i, \phi_i) \rightarrow (\gamma_i, f_{\mathrm{filter}}, \psi)$$

**Finally, the workflow executes as a privacy-aware sequence over all nodes:**

$$O_{\mathrm{final}}, \mathcal{E}_{\mathrm{final}} = \prod_{\substack{v_i \in V \\ (v_i, v_{i+1}) \in E}} \mathrm{LLM}_{\mathrm{Executor}}(\gamma_i, q_i, \mathcal{E}_i, O_i, \phi_i)$$

This distributed design ensures agentic workflow is planned to manage both **task utility** and **privacy preservation**.

# PrivAgentFlow: Workflow Execution with Distributed Privacy Control

## Experimental Results:

- **PrivAgentFlow** exhibits **performance surpassing** the baseline ***AgentDAM,*** a reasoning baseline.
- **PrivAgentFlow** demonstrates **a significantly lower privacy risk** on the web agent benchmark.
- By combining an adaptive workflow and an intelligent tiered privacy strategy, **PrivAgentFlow optimizes the total cost of ownership** while **guaranteeing privacy.**

| LLM | Number of Parameters | AGENTDAM (Baseline) | | AGENTDAM + PrivacyCoT | | PrivAgentFlow (Ours) | |
|---|---|---|---|---|---|---|---|
| | | util (↑) | priv (↓) | util (↑) | priv (↓) | util (↑) | priv (↓) |
| gpt-4o | 200B | 0.655 | 0.167 | $0.643_{\downarrow 0.012}$ | $0.095_{\downarrow 0.072}$ | $0.667_{\uparrow 0.012}$ | $\mathbf{0.012}_{\downarrow 0.155}$ |
| gpt-4o-mini | 8B | 0.631 | 0.071 | $0.595_{\downarrow 0.036}$ | $0.119_{\uparrow 0.048}$ | $0.643_{\uparrow 0.012}$ | $\mathbf{0.012}_{\downarrow 0.059}$ |
| gpt-4-turbo | 20B | 0.667 | 0.179 | $0.643_{\downarrow 0.024}$ | $0.107_{\downarrow 0.072}$ | $0.619_{\downarrow 0.048}$ | $\mathbf{0.048}_{\downarrow 0.131}$ |
| llama-3.3-70b | 70B | 0.667 | 0.083 | 0.667 | $0.048_{\downarrow 0.035}$ | $0.690_{\uparrow 0.023}$ | $\mathbf{0.059}_{\downarrow 0.024}$ |

# PrivAgentFlow: Workflow Execution with Distributed Privacy Control

## Conclusion:

- Trustworthy autonomy arises not from stronger models alone, but from **principled agentic architectures** that make autonomy controllable and auditable.

- PrivAgentFlow embeds privacy and governance directly into agentic decision flows, transforming privacy from a passive filter into an **active control constraint**.

- Workflow-grounded execution makes agent behavior **transparent, auditable, and policy-compliant** — a prerequisite for trustworthy autonomy.

- Distributed node-level control enables **governable autonomy** without sacrificing accountability, safety, or user trust.

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